

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

III Semester of M. Sc. (Biotechnology/Microbiology/Biochemistry) Examination, November 2019

Course code & Name: MS 811 & Omics and Bioinformatics

Date: 06/11/2019 Day: Wednesday Time: 1:30 P.M. To 2:00 P.M. Maximum Marks: 20

MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
- Use of non-programmable calculator is allowed.

Q - I	Choose the correct answer for the following questions.		20
1.	The following genome sequencing technology is based on detection of fluorescence.		01
	(i) Ion-torrent	(ii) SOLiD method	
	(iii) Oxford nanopore	(iv) All of them	
2.	In order to construct a shotgun genomic library of the human genome with an average insert size of 150 Kb. What will be the number of clones that you will require for a four-fold coverage of the entire genome sequence?		01
	(i) 99000 clones	(ii) 88000 clones	
	(iii) 66000 clones	(iv) 44000 clones	
3.	The following method utilizes phosphoramidite chemistry and optical systems to synthesize oligonucleotides directly on surface of a glass chips for manufacturing DNA microarrays		01
	(i) Non-contact printing	(ii) Contact printing	
	(iii) Photolithography	(iv) All of them	
4.	The _____ genome sequencing technology, utilizes nanocontainers (Zero Mode Waveguides) to allow real-time detection of fluorescence generated upon single nucleotide incorporations by DNA polymerases.		01
	(i) Illumina-HiSeq	(ii) MinION	

	(iii) PacBio SMRT	(iv) Semiconductor	
5.	The following transcriptomics technique can accurately delineate the beginning and end of mRNA transcripts; reveal post-transcriptional modifications such as intron splice-variants.		01
	(i) SAGE	(ii) DNA microarrays	
	(iii) RNASeq	(iv) MudPIT	
6.	Which of the following quantitative proteomics method utilizes light and heavy versions of biotinylated iodoacetamide reagent to label the proteome samples?		01
	(i) Protein microarrays	(ii) Isotope coded affinity tag	
	(iii) Metabolic stable isotope coding	(iv) Enzymatic stable isotope coding	
7.	In eukaryotic genome organization, the following sequences can be classified as tandem repeats.		01
	(i) LINES	(ii) Retroposons	
	(iii) DNA transposons	(iv) Microsatellite	
8.	In yeast-two hybrid system, the prey library is constructed by cloning cDNA sequences into the vector adjacent to the DNA sequence for the _____ domain of the transcription factor Gal4 and then introduced into yeast cells.		01
	(i) Transcription-activation	(ii) DNA binding	
	(iii) Ribosome-binding	(iv) Calmodulin-binding	
9.	Which of the following accession number format denotes experimentally derived mRNA sequence in the RefSeq database?		01
	(i) NT_123456	(ii) NP_123456	
	(iii) NM_123456	(iv) All of them	
10.	Which of the following is a biochemical method that infers the binary protein-protein interactions by conjugating two proteins with either donor or acceptor fluorophores.		01
	(i) Rosetta stone	(ii) Yeast-two hybrid system	
	(iii) Metabolic stable-isotope coding	(iv) FRET analysis	
11.	Homologous genes arising by gene duplication events within a single organism and encoding proteins with related but non-identical functions are known as _____.		01
	(i) Orthologs	(ii) COGs	
	(iii) Paralogs	(iv) Pseudogenes	
12.	The following is an <i>in vitro</i> library-based screening method allowing the large-scale analysis of binary protein-protein interactions.		01
	(i) Peptide mass fingerprinting	(ii) Yeast two-hybrid system	
	(iii) Phage Display	(iv) Tandem affinity purification	
13.	The following is a distance matrix-based phylogenetic tree building method.		01
	(i) Maximum Likelihood	(ii) Bootstrapping	
	(iii) Maximum Parsimony	(iv) Neighbor Joining	
14.	Which of the following program is not used for validation of proteins structures?		01
	(i) PROCHECK	(ii) DSSP	
	(iii) PHYLIP	(iv) VADAR	
15.	The sandwich-style antibody protein microarrays are used for detection of _____.		01
	(i) Peptide mass-fingerprints	(ii) His-tagged proteins	

	(iii) Streptavidin-labelled lectins	(iv) Phosphorylated proteins	
16.	In phylogenetic analysis the _____ evolutionary model computes distances based on the assumption of equal probability for all substitutions (transitions and transversions).		01
	(i) Bootstrapping	(ii) Jukes-Cantor	
	(iii) UPGMA	(iv) Kimura	
17.	During the isoelectric focusing (IEF), a protein in a pH region above its isoelectric point (pI) will migrate towards the _____ on an immobilized pH gradient (IPG) strip.		01
	(i) Center of the gel	(ii) Anode	
	(iii) Cathode	(iv) Stop migrating	
18.	The following is a quantitative proteomics method in which cells are grown from two distinct cultures in a medium supplemented with either normal amino acids or stable isotope ($^2\text{H}/^{13}\text{C}/^{15}\text{N}$) labeled amino acids.		01
	(i) ICAT	(ii) Enzymatic stable-isotope coding	
	(iii) 2D-DIGE	(iv) Metabolic stable isotope coding	
19.	The average distance between the atoms of superimposed proteins is expressed as root mean square deviation (RMSD) in which of the following protein structure comparison methods?		01
	(i) Point accepted mutation (PAM)	(ii) Intermolecular method	
	(iii) BLOSUM	(iv) Intramolecular method	
20.	Which of the following program is used for assembly of genome sequences from short next-generation sequencing (NGS) reads?		01
	(i) Velvet	(ii) SOAP	
	(iii) ABySS	(iv) All of them	

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

III Semester of M. Sc. (Biotechnology/Microbiology/Biochemistry) Examination, November 2019

Course code & Name: MS 811 & Omics and Bioinformatics

Date: 06/11/2019 Day: Wednesday Time: 2:00 P.M. To 4:30 P.M. Maximum Marks: 50

Instructions:

1. Section I and II must be attempted in TWO ANSWER SHEET.
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of non-programmable calculator is allowed.
4. Show necessary calculations.

SECTION - I		
Q – II Answer the following questions as directed (answer any five).		20
1.	Explain how a DNA microarray experiment can be performed for differential gene expression analysis in tissue samples obtained from healthy and diseased individuals?	04
2.	Explain the high-throughput genome sequencing using Sequencing by Oligonucleotide Ligation and Detection (SOLiD) technology.	04
3.	Give an overview of pharmacogenomics and explain citing suitable examples, how genotype variations affect the clinical outcomes of drugs in different individuals.	04
4.	Explain the high-throughput Illumina/Solexa genome sequencing technology.	04
5.	Explain the genome sequence assembly using the de Bruijn graph method.	04
6.	Explain the various steps in the whole genome shotgun sequencing approach.	04
7.	Describe the different categories of repetitive DNA present in the eukaryotic genomes.	04
SECTION - II		
Q-III Answer the following questions as directed (answer any ten).		30
1.	Explain the detection of posttranslational protein glycosylation using protein microarrays	03
2.	Explain the quantitative proteome analysis using enzymatic stable isotope coding.	03
3.	Explain the whole proteome wide screening of binary protein-protein interactions using yeast two hybrid systems.	03
4.	Give a detailed workflow of 2D-PAGE and identification of separated protein spots by peptide mass fingerprinting.	03
5.	Explain the Rosetta stone method for prediction of protein-protein interactions.	03
6.	Explain the multidimensional protein identification technology (MudPIT).	03
7.	Explain quantitative proteome expression analysis using isotope-coded affinity tags.	03
8.	Explain the UPGMA method for phylogenetic tree construction using a suitable example.	03
9.	Enlist the criteria for a good protein structure.	03
10.	What is a biological database? Describe giving suitable examples the different types of biological databases.	03
11.	Describe the key differences between global and local sequence alignments.	03
12.	Expand the following terms: (i) PAM (ii) MEGA (iii) SCOP	03